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Chapter-1: Introduction to Natural Language Processing

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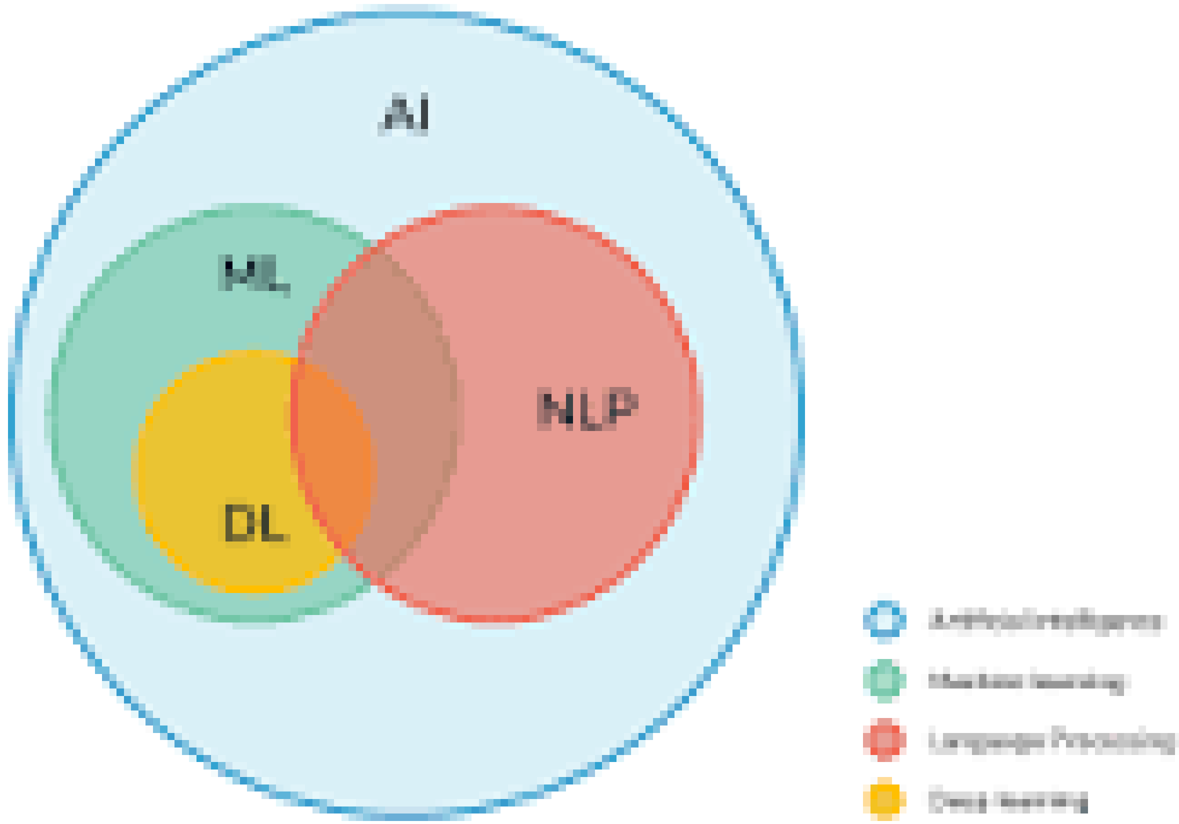
Basic Information About DL and NLP



The First Wave: Rationalism

The Second Wave: Empiricism

The Third Wave: Deep Learning

Rationalism:

1. **Rationalism:** Rationalism is a philosophical concept that emphasizes the role of reason and logical thinking in acquiring knowledge.

- It argues that knowledge is derived from innate ideas or through the use of deductive reasoning.

- In the context of NLP, rationalism may refer to the development of rule-based systems or symbolic approaches to language understanding, where explicit rules and logic are used to process and analyze natural language.

Empiricism

2. Empiricism: Empiricism is a philosophical concept that emphasizes the role of experience and sensory perception in acquiring knowledge.

- It <u>suggests</u> that knowledge is gained through observation, experimentation, and empirical evidence.

- In the context of NLP, empiricism may refer to approaches that rely on large amounts of annotated data and statistical modeling to learn patterns and extract information from natural language.

Empiricism

3. Deep Learning: Deep learning is a subfield of machine learning that focuses on the development and application of artificial neural networks with multiple layers (hence the term "deep").

- Deep learning models, known as deep neural networks, have the ability to automatically learn hierarchical representations of data through training on large datasets.

- In the context of NLP, deep learning has made significant advancements in tasks such as language modeling, sentiment analysis, machine translation, and text generation.

NLP- Natural Language Processing

What is NLP?

- Natural language processing (NLP) is a subfield of computer science and artificial intelligence (AI) that uses machine learning to enable computers to understand and communicate with human language.
- Natural Language Processing (NLP) is a branch of artificial intelligence (AI) that focuses on the interaction between computers and human language.
- Its primary goal is to enable machines to understand, interpret, and generate natural language in a way that is meaningful and useful.

NLP- Natural Language Processing

NLP enables computers and digital devices to recognize, understand and generate text and speech by combining computational linguistics.

The rule-based modelling of human language—together with statistical modelling, machine learning (ML) and deep learning.

NLP- Natural Language Processing

- NLP research has enabled the era of generative AI, from the communication skills of large language models (LLMs) to the ability of image generation models to understand requests.

- NLP is already part of everyday life for many, powering search engines, prompting chatbots for customer service with spoken commands, voice-operated GPS systems and digital assistants on smartphones.

NLP- Natural Language Processing

- A natural language processing system can work rapidly and efficiently: after NLP models are properly trained, it can take on administrative tasks, freeing staff for more productive work. Benefits can include-

- **Faster insight discovery**: Organizations can find hidden patterns, trends and relationships between different pieces of content. Text data retrieval supports deeper insights and analysis, enabling better-informed decision-making and surfacing new business ideas.

NLP- Natural Language Processing

- **Greater budget savings**: With the massive volume of unstructured text data available, NLP can be used to automate the gathering, processing and organization of information with less manual effort.
- **Quick access to corporate data**: An enterprise can build a knowledge base of organizational information to be efficiently accessed with AI search.
- For sales representatives, NLP can help quickly return relevant information, to improve customer service and help close sales.

NLP- Natural Language Processing

Challenges of NLP

- NLP models are not perfect and probably never will be, just as human speech is prone to error. Risks might include:
- **Biased training**: As with any AI function, biased data used in training will skew the answers. The more diverse the users of an NLP function, the more significant this risk becomes, such as in government services, healthcare and HR interactions. Training datasets scraped from the web, for example, are prone to bias.

NLP- Natural Language Processing

- **Mis-interpretation**: As in programming, there is a risk of garbage in, garbage out (GIGO). NLP solutions might become confused if spoken input is in an obscure dialect, mumbled, too full of slang, homonyms, incorrect grammar, idioms, fragments, mispronunciations, contractions or recorded with too much background noise.

NLP- Natural Language Processing

- **New vocabulary:** New words are continually being invented or imported. The conventions of grammar can evolve or be intentionally broken. In these cases, NLP can either make a best guess or admit it's unsure—and either way, this creates a complication.

- **Tone of voice**: When people speak, their verbal delivery or even body language can give an entirely different meaning than the words alone. overemphasis for effect, stressing words for importance or sarcasm can be confused by NLP, making the semantic analysis more difficult and less reliable.

NLP- Natural Language Processing

How NLP works?

- Human language is filled with many ambiguities that make it difficult for programmers to write software that accurately determines the intended meaning of text or voice data.
- Programmers must teach natural language-driven applications to recognize and understand irregularities so their applications can be accurate and useful.

NLP- Natural Language Processing

- NLP combines the power of computational linguistics together with machine learning algorithms and deep learning. Computational linguistics is a discipline of linguistics that uses data science to analyse language and speech.

NLP- Natural Language Processing

- **Why is NLP important?**

- Natural language processing (NLP) is critical to fully and efficiently analyse text and speech data. It can work through the differences in dialects, slang, and grammatical irregularities typical in day-to-day conversations.

- **Companies use it for several automated tasks, such as to:**
  - Process, analyse, and archive large documents
  - Analyse customer feedback or call centre recordings
  - Run chatbots for automated customer service
  - Answer who-what-when-where questions
  - Classify and extract text

What are NLP use cases for business?

- Businesses use natural language processing (NLP) software and tools to simplify, automate, and streamline operations efficiently and accurately. We give some example use cases below.

- Sensitive data redaction

- Businesses in the insurance, legal and healthcare sectors process, sort, and retrieve large volumes of sensitive documents like medical records, financial data, and private data.

- Instead of reviewing manually, companies use NLP technology to redact personally identifiable information and protect sensitive data.

Customer Engagement

- NLP technologies allow chat and voice bots to be more human-like when conversing with customers. Businesses use chatbots to scale customer service capability and quality while keeping operational costs to a minimum.
- PubNub, which builds chatbot software, uses Amazon Comprehend to introduce localized chat functionality for its global customers.
- T-Mobile uses NLP to identify specific keywords in customers' text messages and offer personalized recommendations.
- Q&A chatbot solution to address student questions using machine learning technology.

Business Analytics

- Marketers use NLP tools like Amazon Comprehend and Amazon Lex to gain an educated perception of what customers feel toward a company's product or services.
- By scanning for specific phrases, they can gauge the customers' moods and emotions in written feedback.
- For example, Success KPI provides natural language processing solutions that help businesses focus on targeted areas in sentiment analysis and help contact centres derive actionable insights from call analytics.

NLP- Natural Language Processing

It includes two main types of analysis:

(1) Syntactical analysis - Syntactical analysis determines the meaning of a word, phrase or sentence by parsing the syntax of the words and applying preprogrammed rules of grammar.

(2) Semantical analysis uses the syntactic output to draw meaning from the words and interpret their meaning within the sentence structure.

What are NLP tasks?

- Natural language processing (NLP) techniques or NLP tasks, break down human text or speech into smaller parts that computer programs can easily understand. Common text processing and analysing capabilities in NLP are given below.
- Part-of-speech tagging
- This is a process where NLP software tags individual words in a sentence according to contextual usages, such as nouns, verbs, adjectives, or adverbs. It helps the computer understand how words form meaningful relationships with each other.

- It helps developers to organize knowledge for performing tasks such as-
- **Translation,**
- **Automatic Summarization,**
- **Named Entity Recognition (NER)**
- **Speech Recognition**
- **Relationship Extraction**
- **Topic Segmentation**

The Goal of NLP

- The goal of NLP is to bridge the gap between human language and machine understanding, enabling computers to process, analyze, and generate text or speech.

NLP encompasses a wide range of tasks and applications,

including but not limited to:

1. **Text Classification:** Categorizing text documents into predefined categories or classes, such as sentiment analysis, spam detection, topic classification, etc.
2. **2. Named Entity Recognition (NER):** Identifying and extracting specific named entities from text, such as names of people, organizations, locations, dates, etc.

The Goal of NLP

3. **Sentiment Analysis:** Determining the sentiment or opinion expressed in a piece of text, whether it is positive, negative, or neutral.

4. **Machine Translation**: Automatically translating text or speech from one language to another, such as Google Translate.

5. **Question Answering:** Building systems that can understand and answer questions posed in natural language, like virtual assistants or chatbots.

6. **Text Summarization:** Generating concise summaries of long texts or documents, capturing the most important information.

The Goal of NLP

7. **Natural Language Generation (NLG):** Creating human-like text or speech output from structured data or other forms of input, such as generating reports or writing news articles.

8. **Language Modeling:** Developing statistical models that capture the structure and patterns of language, often used in tasks like speech recognition, machine translation, and text generation.

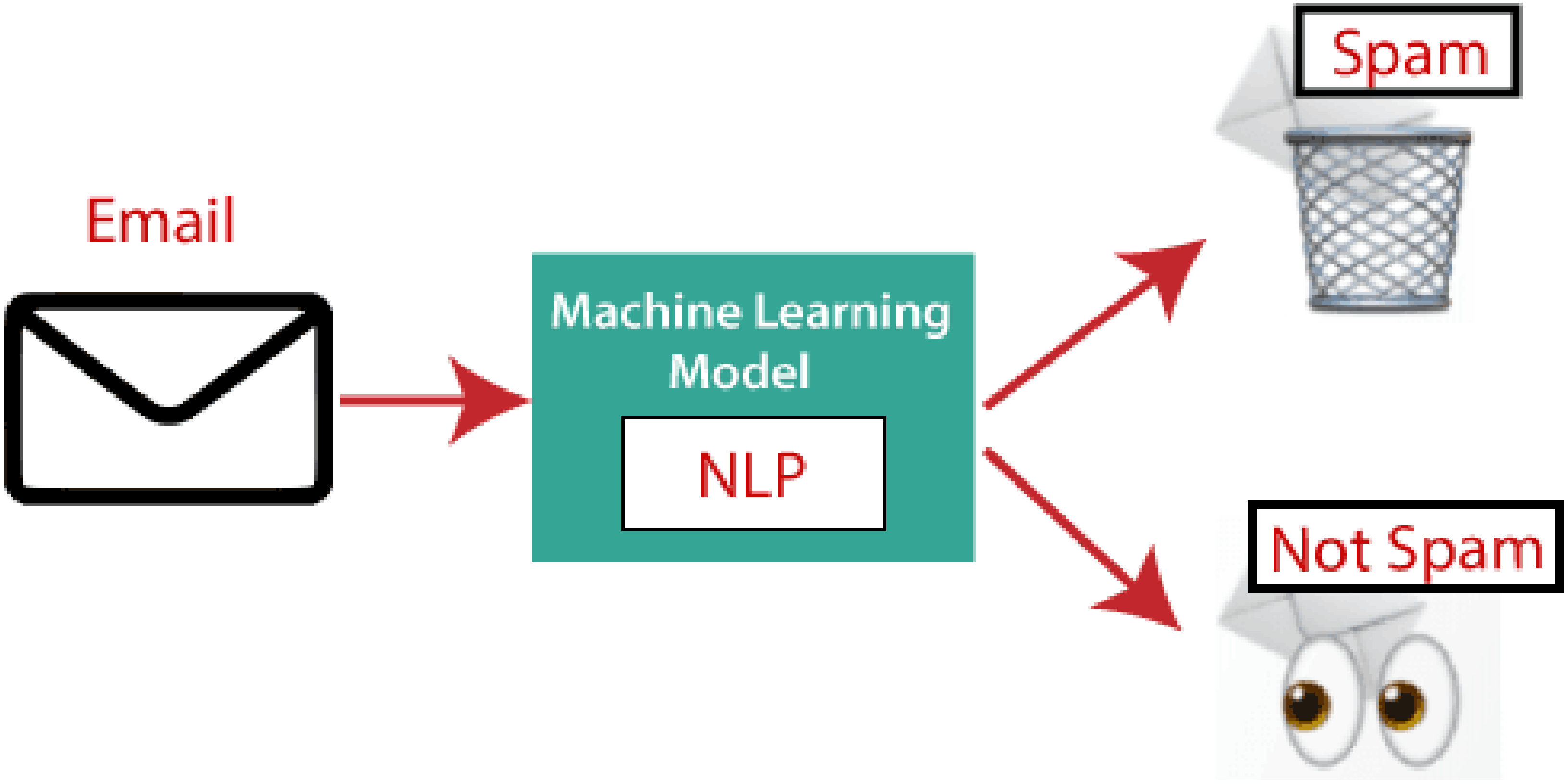
- Natural Language Processing (NLP) involves a range of operations to understand and process human language. Here's an **introduction to three fundamental aspects of NLP operations**:
- **Word Level Analysis**: Word level analysis in NLP focuses on understanding individual words within a given text. This involves tasks such as tokenization, stemming, lemmatization, and part-of-speech (POS) tagging.
- **Stemming and lemmatization simplify words into their root form. For example, these processes turn "starting" into "start."**
- **Stop word removal ensures that words that do not add significant meaning to a sentence, such as "for" and "with," are removed.**

Part-of-speech (POS) tagging: Assigning a grammatical category (noun, verb, adjective, etc.) to each word in a sentence.

Applications of NLP

1. **Question Answering:** Question Answering focuses on building systems that automatically answer the questions asked by humans in a natural language.

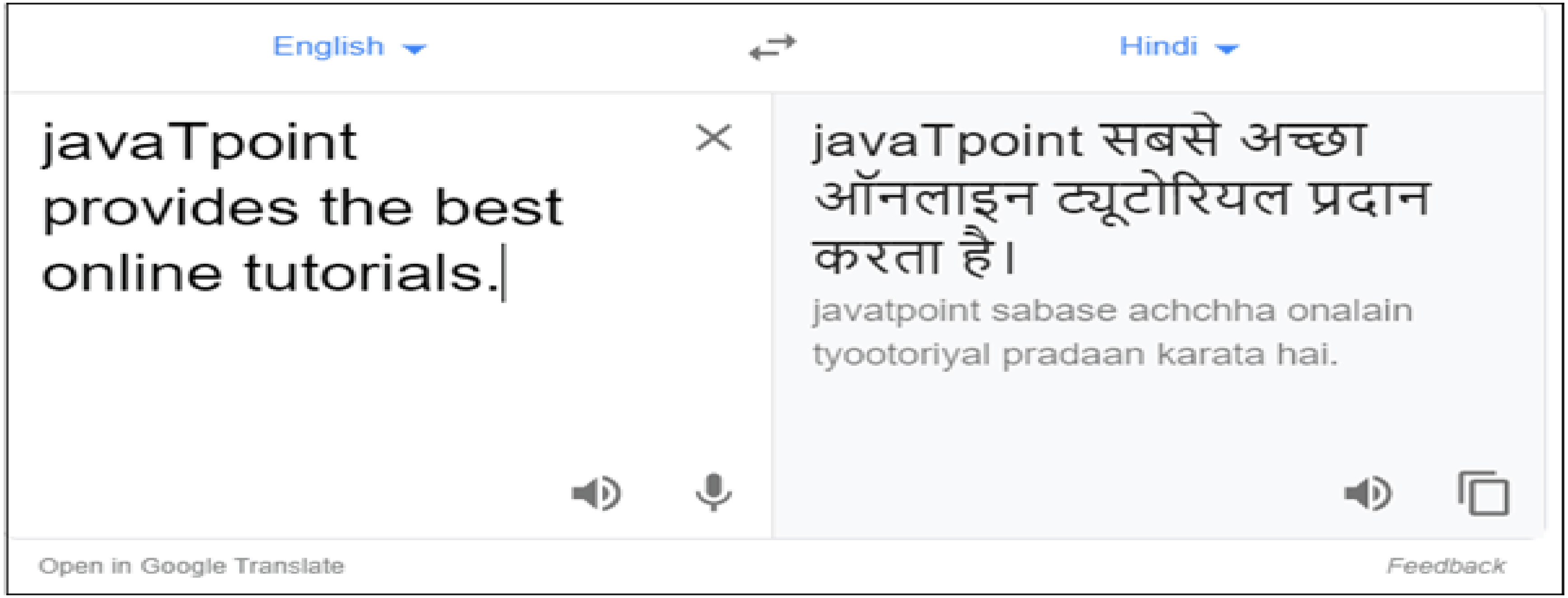
2. Spam Detection: Spam detection is used to detect unwanted e-mails getting to a user's inbox.



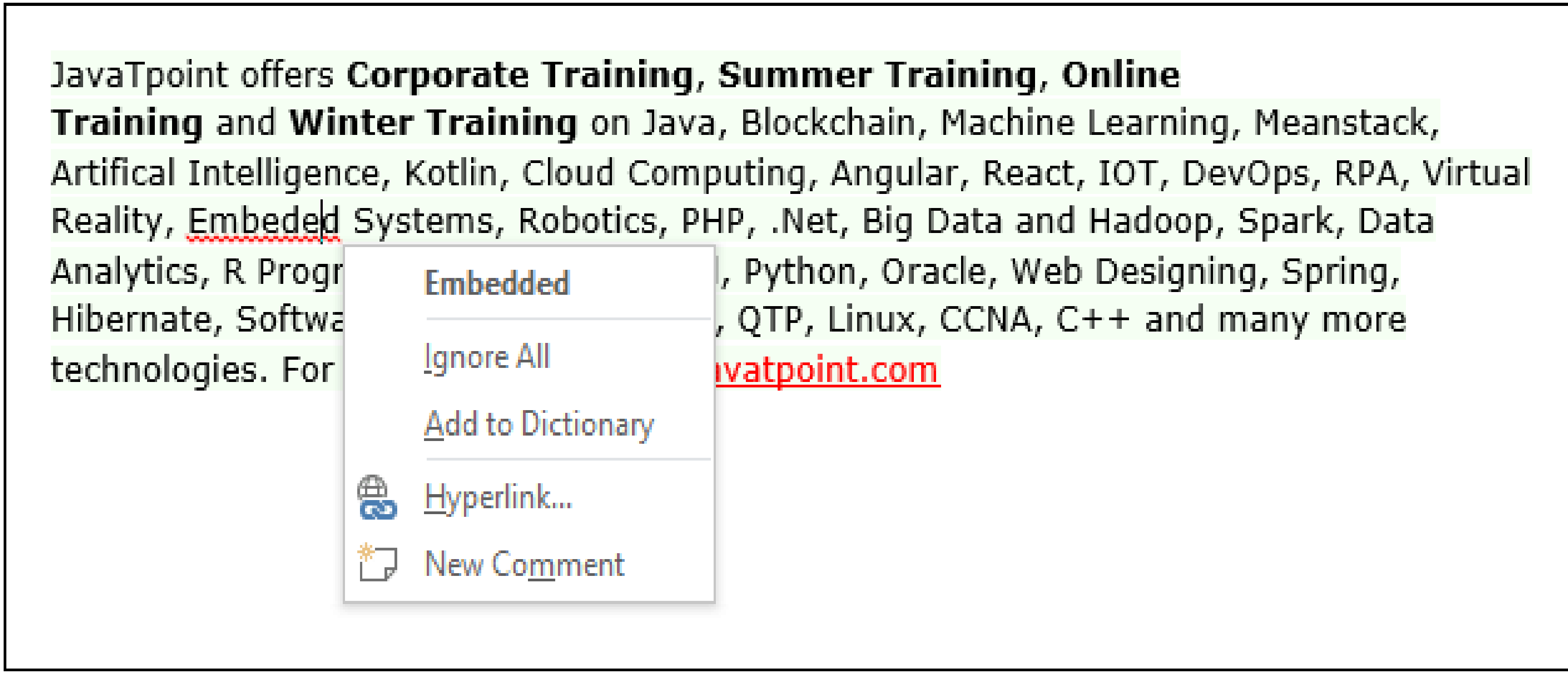
3. Sentiment Analysis: Sentiment Analysis is also known as **opinion mining**. It is used on the web to analyse the attitude, behaviour, and emotional state of the sender.

4. Machine Translation: Machine translation is used to translate text or speech from one natural language to another natural language.

Example: Google Translator.



5. **Spelling correction:** Microsoft Corporation provides word processor software like MS-word, PowerPoint for the spelling correction.

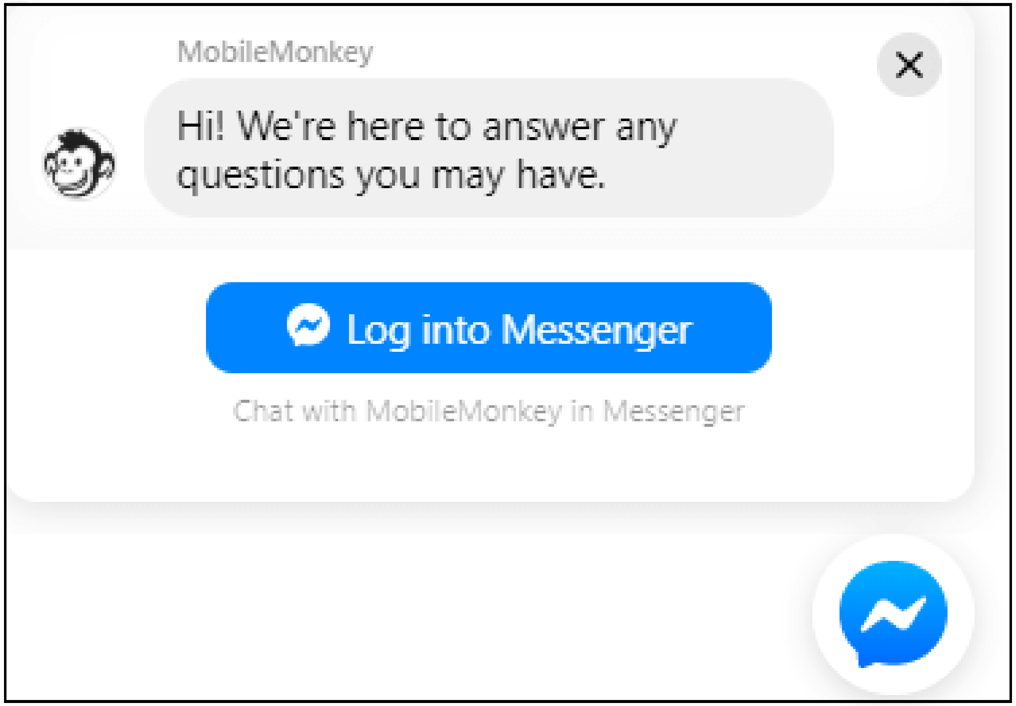


6. **Speech Recognition:** Speech recognition is used for converting spoken words into text. It is used in applications, such as mobile, home automation, video recovery, dictating to Microsoft Word, voice biometrics, voice user interface, and so on.

7. **Information extraction:** Information extraction is one of the most important applications of NLP. It is used for extracting structured information from unstructured or semi-structured machine-readable documents.

8. **Natural Language Understanding (NLU):** It converts a large set of text into more formal representations such as first-order logic structures that are easier for the computer programs to manipulate notations of the natural language processing.

9. Chatbot: Implementing the Chatbot is one of the important applications of NLP. It is used by many companies to provide the customer's chat services.



The main components of NLP

- **Tokenization:** Breaking down text into smaller units, such as words or sentences, known as tokens. Tokenization is a fundamental step in NLP as it provides a basis for further analysis.

- **Morphological Analysis:** Understanding the structure and form of words, including identifying their roots, prefixes, suffixes, and inflections. This analysis helps in tasks such as stemming or lemmatization.

- **Syntax Analysis:** Analyzing the grammatical structure of sentences to determine the relationships between words. This includes tasks like part-of-speech tagging, parsing, and syntactic analysis.

- **Semantics:** Understanding the meaning of words, phrases, and sentences. This involves tasks such as word sense disambiguation, semantic role labeling, and semantic parsing.

- **Discourse Analysis:** Analyzing how sentences and ideas are connected to form coherent and cohesive texts. It involves tasks like co reference resolution, discourse parsing, and text coherence analysis.

What are Corpus, Tokens and Engrams?

- A **Corpus** is defined as a collection of text documents.

- For example a data set containing news is a corpus or the tweets containing Twitter data is a corpus.

- So corpus consists of documents, documents comprise paragraphs, paragraphs comprise sentences and sentences comprise further smaller units which are called **Tokens**.

➢ Corpus>Documents>Paragraphs>Sentence>Tokens

- Tokens can be words, phrases, or Engrams, and **Engrams** are defined as the group of n words together.
- The Uni-grams are representing one word,
- Di-grams are representing two words together and
- Tri-grams are representing three words together.

- For example, consider this given sentence-

"I love my phone."

- In this sentence, the uni-grams(n=1) are:

I, love, my, phone

- Di-grams (n=2) are: **I love, love my, my phone**

- Tri-grams (n=3) are: **I love my, love my phone**

What is Tokenization?

- Tokenization is a process of splitting a text object into smaller units which are also called tokens.
- Examples of tokens can be words, numbers, engrams, or even symbols.
- The most commonly used tokenization process is **White-space Tokenization**.

What is White-space Tokenization?

- Also known as unigram tokenization. In this process, the entire text is split into words by splitting them from white spaces.

- For example, in a sentence- “I went to New-York to play football.”

- This will be splitted into following tokens: “I”, “went”, “to”, “New-York”, “to”, “play”, “football.”

- **Notice** that “New-York” is not split further because the tokenization process was based on whitespaces only.

What is Regular Expression Tokenization?

- The other type of tokenization process is **Regular Expression Tokenization,** in which a regular expression pattern is used to get the tokens.
- For example, consider the following string containing multiple delimiters such as comma, semi-colon, and white space.
- Sentence= “Football, Cricket; Golf Tennis",
- Sentence
- Tokens= “Football”, ”Cricket”, “Golf”, “Tennis”

Remarks*-

- **Tokenization can be performed at the sentence level or at the world level or even at the character level.**

What is Normalization?

Before normalization we read- **Morpheme**,

Morpheme is defined as the base form of a word.

- A token is generally made up of two components,

1. Morphemes- which are the base form of the word,
2. Inflectional - which are essentially the suffixes and prefixes added to morphemes.

Structure of Tokens: <prefix><morpheme><suffix>

For example, consider the word "**Antinationalist",**

Example: Antinationalist : Anti + national + ist

which is made up of **Anti** and **ist** as the inflectional forms

and **national** as the morpheme.

- **Normalization:** Normalization is the process of converting a token into its base form.

- In the normalization process, the inflection from a word is removed so that the base form can be obtained.

- So, the normalized form of anti-nationalist is national.

- Normalization is useful in reducing the number of unique tokens present in the text, removing the variations of a word in the text, and removing redundant information too.

- Popular methods which are used for normalization are Stemming and Lemmatization.

What is Stemming?

- Stemming is an elementary rule-based process for removing inflectional forms from a token and the outputs are the **stem** of the world.
- For example,
- “laughing”,
- “laughed“,
- “laughs”,
- “laugh”
- will all become “laugh”,

which is their stem, because their inflection form will be removed.

- Stemming is <u>not</u> a good normalization process because sometimes stemming can produce words that are not in the dictionary.

For example, consider a sentence: "His teams are not winning".

- After stemming the tokens that we will get are- "hi", "team", "are", "not", "winn"

- Notice that the keyword **"winn"** is not a regular word and

- "**hi**" changed the context of the entire sentence.

- Another example could be-

Form	Suffix	Stem
Studies	-es	Studi
Studying	-ing	Study

What is Lemmatization?

- Lemmatization, on the other hand, is a systematic step-by-step process for removing inflection forms of a word.
- It makes use of vocabulary, word structure, part of speech tags, and grammar relations.
- The output of lemmatization is the root word called **a lemma**.

For example-

1. Am, Are, Is >> Be
2. Running, Ran, Run >> Run

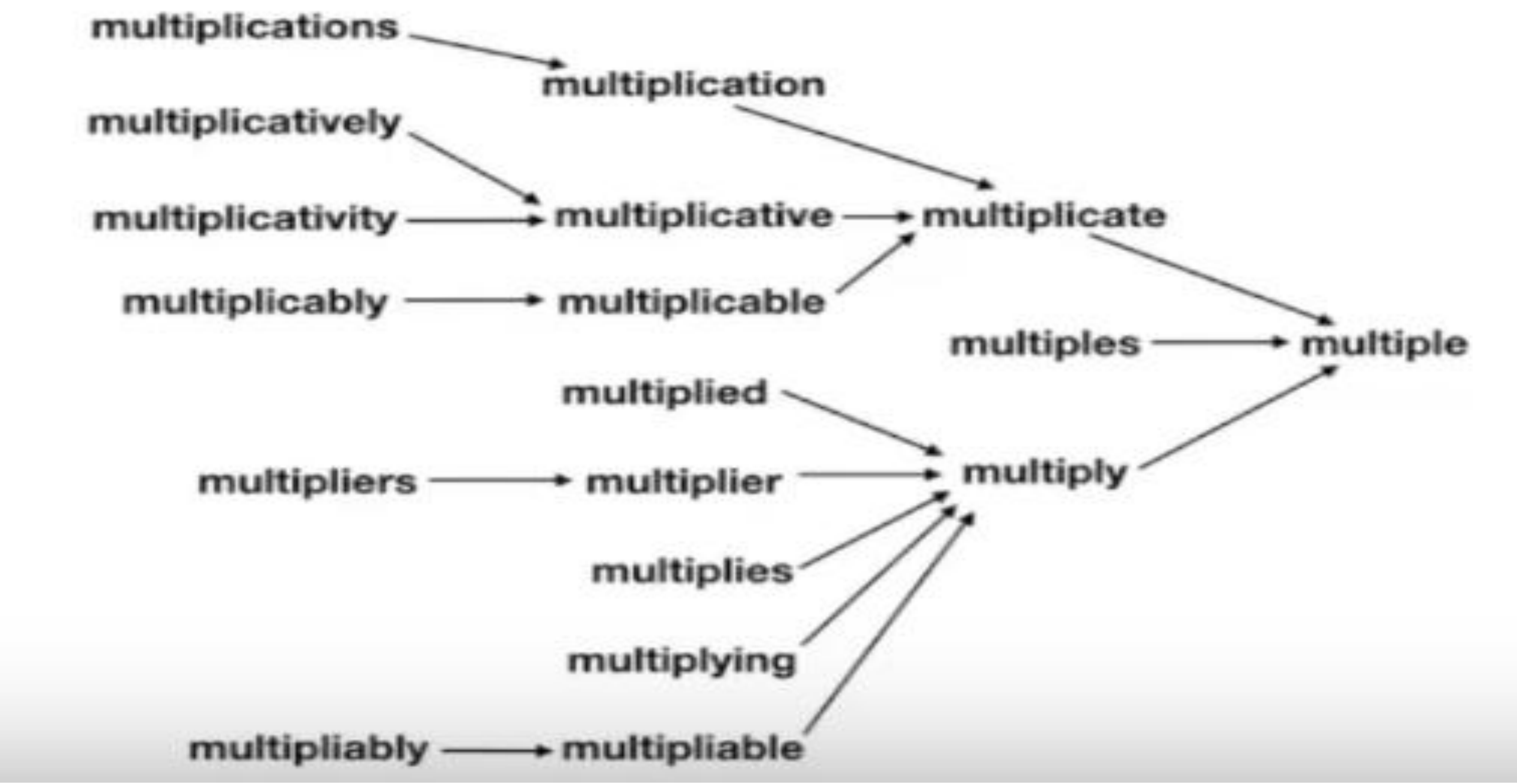
- Also, since it is a systematic process while performing lemmatization one can specify the part of the speech tag for the desired term and lemmatization will only be performed if the given word has the proper part of the speech tag.

- For example

- if we try to lemmatize the word **running** as a **verb**, it will be converted to **run.**

- But if we try to lemmatize the same word **running** as a **noun** it won't be converted.

A detailed explanation of how Lemmatization works by the step-by-step process to remove inflection forms of a word-



Part of Speech (PoS) Tags in Natural Language Processing-

- Part of Speech or PoS tags is the properties of words that define their main **context, their function and the usage** in a sentence.
- Some of the commonly used parts of speech tags are

  Nouns, which define any object or entity;

  Verbs, which define some action; and

  Adjectives or **Adverbs**, which act as the modifiers, quantifiers, or intensifiers in any sentence.

- In a sentence, every word will be associated with a proper part of the speech tag.

for example-

“David has purchased a new laptop from the Apple store.”

- In the below sentence, every word is associated with a part of the speech tag which defines their functions.

- In this case-
- “David’ has **NNP** tag which means it is a proper <u>noun</u>,
- “has” and “purchased” belongs to <u>verb</u> indicating that they are the actions and
- “laptop” and “Apple store” are the <u>nouns</u>,
- “new” is the <u>adjective</u> whose role is to modify the context of laptop.

- Part of speech tags is defined by the <u>relations of words</u> with the <u>other words</u> in the sentence.

- Machine learning models or rule-based models are applied to obtain the part of speech tags of a word.

- Part of speech tags have a large number of applications and they are used in a variety of tasks such as
- **text cleaning,**
- **feature engineering tasks and**
- **word sense disambiguation**.

- For example, consider these two sentences-

Sentence1: "Please **book** my flight for New-York"

Sentence2: "I like to read a **book** on New-York"

  - In both sentences, the keyword "**book**" is used but In sentence one, it is used as a **verb** while in

    sentence two it is used as a **noun**.

What makes Natural Language Processing Difficult?

Natural Language Processing (NLP) poses several challenges that make it a complex and difficult task. Here are some reasons why NLP is challenging:

1. **Ambiguity:** Natural language is inherently ambiguous.

- Words or phrases can have multiple meanings depending on the context in which they are used.
- Resolving this ambiguity and accurately interpreting the intended meaning requires a deep understanding of language and context.

What makes Natural Language Processing Difficult?

2. Contextual Understanding: Understanding language goes beyond the literal meaning of words.

- It involves comprehending the underlying context, implicit meanings, idiomatic expressions, metaphors, sarcasm, and cultural references.

- Incorporating contextual understanding into NLP systems requires sophisticated models and techniques.

What makes Natural Language Processing Difficult?

3. Variability and Creativity: Human language is highly variable and constantly evolving.

- People use different dialects, accents, slang, and even create new words or phrases.
- NLP systems must be robust enough to handle this variability and adapt to new linguistic patterns.

4. Data Sparsity: While human language has infinite possibilities, the available labeled training data for NLP tasks is often limited.

- Annotating large datasets with human-labeled data is expensive and time-consuming.
- This scarcity of labeled data makes it challenging to build accurate and generalizable NLP models.

What makes Natural Language Processing Difficult?

5. **Named Entity Recognition (NER):** Identifying named entities, such as names of people, organizations, or locations, is a crucial task in NLP.

- However, the wide variety of names, variations, misspellings, and the constant addition of new entities make it a difficult challenge.

6. **Syntax and Grammar:** Language follows grammatical rules and syntactic structures, which play a significant role in understanding its meaning.

- However, human language is not always grammatically correct.
- It can have errors, incomplete sentences, or deviations from formal grammar rules, making it challenging for NLP systems to parse and understand.

What makes Natural Language Processing Difficult?

7. Domain and Context Adaptation: NLP models trained on one domain or context often struggle to generalize to other domains or adapt to new contexts.

- The ability to transfer knowledge from one domain to another or adapt to different contexts remains a significant challenge in NLP.

8. Cultural and Linguistic Diversity: Language varies across different cultures, regions, and languages.

- NLP systems need to account for this diversity and ensure fairness and inclusivity in their analysis and processing.

Achieve Through Natural Language Processing

The overarching goal of Natural Language Processing (NLP) is to enable computers to understand, interpret, and generate human language in a way that is both meaningful and useful. Here are some specific objectives and achievements sought through NLP:

1. **Language Understanding:** NLP aims to develop models and algorithms that can comprehend and understand human language in a manner similar to how humans do.

- This involves tasks such as part-of-speech tagging, syntactic parsing, and semantic analysis to extract information about the structure and meaning of text or speech.

2. **Language Generation:** NLP models can generate human-like text based on given prompts or conditions. This includes tasks like generating coherent and contextually relevant responses, text synthesis, and creative writing.

Achieve Through Natural Language Processing

3. **Information Retrieval:** NLP helps in effectively retrieving relevant information from large volumes of textual data. This includes tasks like keyword extraction, document classification, and information extraction, enabling users to search and access the desired information more efficiently.

4. **Text Summarization:** NLP can automatically summarize long documents, articles, or text passages, providing condensed versions that capture the key points and main ideas. Text summarization facilitates quick comprehension and efficient information consumption.

Achieve Through Natural Language Processing

5. **Machine Translation**: NLP plays a significant role in machine translation systems, aiming to accurately translate text or speech from one language to another. These systems use statistical models, neural networks, or rule-based approaches to enable communication across different languages.

6. **Question Answering:** NLP facilitates the development of question-answering systems that can comprehend user questions and provide relevant and accurate answers. These systems may leverage knowledge bases, search algorithms, or information retrieval techniques to extract the most appropriate responses.

Achieve Through Natural Language Processing

7. **Sentiment Analysis:** NLP enables the analysis of sentiments and emotions expressed in text or speech. Sentiment analysis models can determine the overall sentiment (positive, negative, neutral) or emotions conveyed in customer reviews, social media posts, or feedback, providing valuable insights for businesses and organizations.

8. **Dialogue Systems:** NLP powers the development of chatbots and virtual assistants capable of engaging in human-like conversations. These systems aim to understand user queries, respond appropriately, and maintain coherent and contextually relevant dialogues.

Common Terms Associated with Language Processing-

In the field of Natural Language Processing (NLP), there are several common terms and concepts that are frequently used. Here are some key terms associated with language processing:

1. **Tokenization:** The process of breaking down a text into smaller units, called tokens, such as words, sentences, or characters. Tokenization is a fundamental step in NLP tasks as it provides the basic units for further analysis.

2. **Part-of-Speech (POS) Tagging:** Assigning grammatical tags to each word in a sentence, indicating its syntactic category or function, such as noun, verb, adjective, etc. POS tagging helps in understanding the grammatical structure of a sentence.

Common Terms Associated with Language Processing-

3. **Lemmatization:** The process of reducing words to their base or dictionary form (lemma). It helps in normalizing words by converting them to a common base form, reducing inflectional variations.

4. **Named Entity Recognition (NER):** Identifying and extracting named entities, such as names of people, organizations, locations, or specific dates, from text. NER is crucial for tasks like information extraction and entity-based analysis.

Common Terms Associated with Language Processing-

5. **Sentiment Analysis:** Analyzing and determining the sentiment or emotion expressed in a given text. Sentiment analysis can categorize text as positive, negative, or neutral, allowing for the assessment of opinions, attitudes, or sentiments.

6. **Text Classification:** Assigning predefined categories or labels to text documents based on their content. Text classification is used in tasks such as spam detection, topic classification, sentiment analysis, and document categorization.

Common Terms Associated with Language Processing-

7. **Language Modeling:** Building probabilistic models that capture the statistical properties of a language, enabling predictions of the likelihood of words or sequences of words. Language models are used in tasks like speech recognition, machine translation, and text generation.

8. **Word Embeddings:** Representing words as numerical vectors in a high-dimensional space, where words with similar meanings are closer to each other. Word embeddings capture semantic relationships and are often used as input for various NLP tasks.

Common Terms Associated with Language Processing-

9. **Dependency Parsing:** Analyzing the grammatical structure of a sentence by determining the dependencies between words. Dependency parsing represents the relationships between words as directed links, capturing how words modify or relate to each other.

10. **Machine Translation:** The task of automatically translating text from one language to another using computational methods. Machine translation systems employ various techniques, including statistical models and neural networks, to generate translations.

NLP Operations

1. **Named Entity Recognition (NER):** Identifying and classifying named entities, such as names of people, organizations, locations, dates, etc., in text.
2. **Sentiment Analysis:** Determining the sentiment or subjective information expressed in text, such as positive, negative, or neutral.
3. **Text Classification:** Categorizing text documents into predefined categories or classes based on their content.
4. **Language Modeling**: Predicting the likelihood of a sequence of words in a given language, which is used in tasks like machine translation and text generation.

NLP Operations

5. **Information Extraction:** Identifying and extracting structured information from unstructured text, such as named entities, relationships, or events.
6. **Text Summarization:** Generating a concise summary of a given text document, either through extractive methods (selecting important sentences) or abstractive methods (generating new sentences).
7. **Machine Translation:** Automatically translating text from one language to another.
8. **Question Answering:** Generating answers to questions based on a given text document or knowledge base.
9. **Topic Modeling:** Identifying the main topics or themes present in a collection of documents.

NLP Operations

10. **Text Generation:** Generating human-like text based on a given prompt or context.
11. **Dependency Parsing:** Analyzing the grammatical relationships between words in a sentence to create a dependency tree.
12. **Coreference Resolution:** Resolving references to entities in text, particularly pronouns and noun phrases.
13. **Relation Extraction:** Identifying and extracting relationships or associations between entities in text.

- These are just a few examples of the NLP operations that are commonly performed. NLP is a vast field with numerous techniques and applications and new advancements continue to expand the range of operations that can be performed on natural language text.

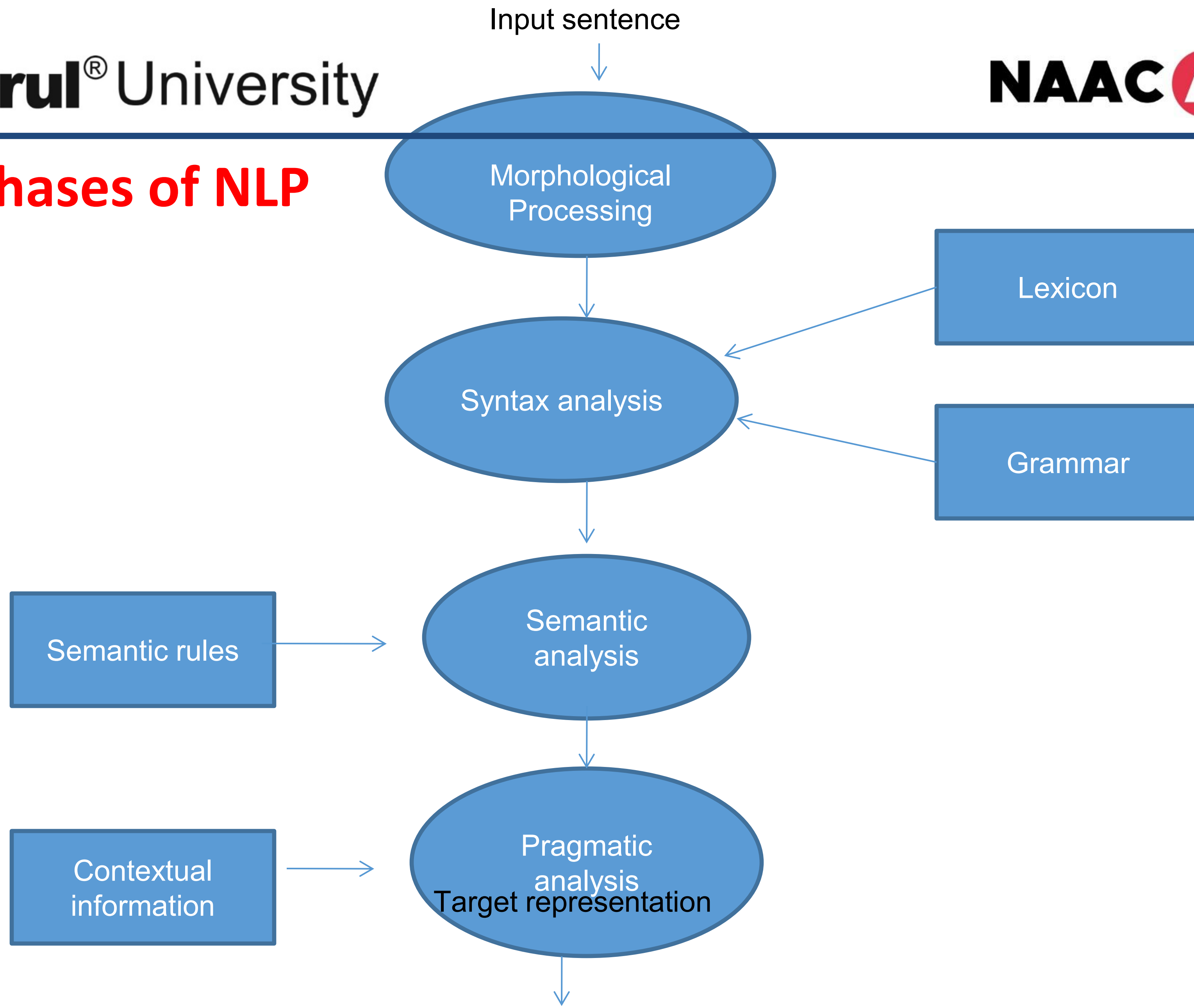
Phases of NLP

1. **Text Preprocessing:** Clean and prepare the raw text data by removing irrelevant characters, lowercasing, handling punctuation, removing stop words, and performing tokenization.

2. **Morphological Processing:** The purpose of this phase is to break chunks of language input into sets of tokens corresponding to paragraphs, sentences and words.

- For example, a word like “uneasy” can be broken into two sub-word tokens as “un-easy”.

Phases of NLP



Phases of NLP

3. Syntax Analysis: The purpose of this phase is two folds: to check that a sentence is well formed or not and to break it up into a structure that shows the syntactic relationships between the different words.

- For example, the sentence like "The school goes to the boy" would be rejected by syntax analyzer or parser.

4. Lexicon: A lexicon is often used to describe the knowledge that a speaker has about the words of a language.

- This includes meanings, use, form, and relationships with other words.
- A lexicon can thus be thought of as a mental dictionary.

Phases of NLP

5. **Semantic analysis:** The purpose of this phase is to draw exact meaning, or you can say dictionary meaning from the text.

- The text is checked for meaningfulness.
- For example, semantic analyzer would reject a sentence like "Hot ice-cream" .

6. Pragmatic Analysis:

- Pragmatic analysis considers the context, intentions, and implications of the text.
- It involves tasks such as speech act recognition, which identifies the speaker's intended action, and dialogue modeling, which deals with natural language understanding and generation in conversational systems.

Phases of NLP

7. Target representation :

- Semantic analysis is concerned with the meaning representation.
- It mainly focuses on the literal meaning of words, phrases, and sentences.

8. Natural Language Generation: Natural Language Generation (NLG) focuses on generating human-like text or responses based on the extracted information or system output.

- It involves tasks like text summarization, machine translation, text-to-speech synthesis, and dialogue generation.

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